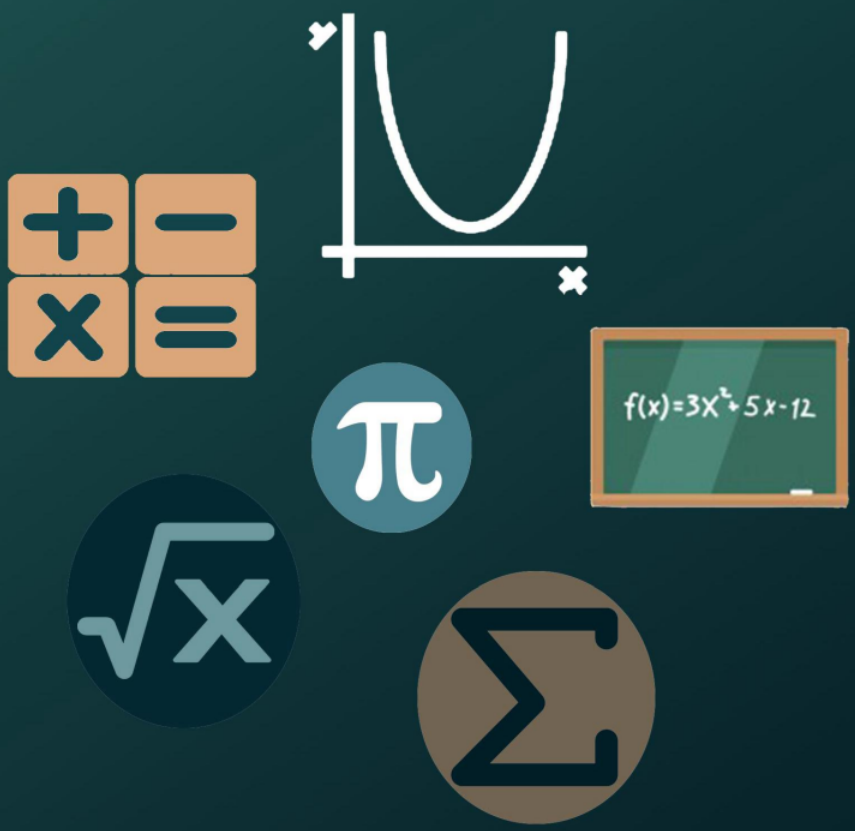
MATHEMATICS

Parabola



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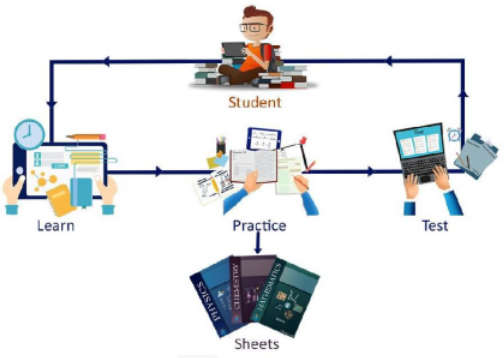
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 5
5.	Exercise 2	Page - 8
6.	Exercise 2A	Page - 11
7.	Answer Key	Page - 14

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Standard Parabola	1-11		1	
2	Shifted parabola	12,13	1	3,4	
3	Point and Parabola	14,15		5-7	1-2
4	Parametric Coordinates of a Parabola	16,17		8	
5	Chord joining two points of a Parabola	18-20	11	9-12	3
6	Length of Chord of Parabola	21-23	12		
7	Tangents to Parabola	24-31		13-15	4-5
8	Pair of Tangents to Parabola	32-39	4-9, 13	16,17	6-7
9	Normals to Parabola	40-46	2-3	18-22	8, 10-12, 16
10	Common Tangents	47	14	23-27	9
11	Subtangent and Subnormal	48,49			
12	Chord with given Mid-Point & Diameter of Parabola	50-53	15	28,29	
13	Highlights about Parabola	54		30-34	
14	After Completing Chapter		10		13-15

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Standard Parabola

Q1. The equation of the directrix of the parabola $y^2 = 12x$ is-

(A) $x + 3 = 0$ (B) $y + 3 = 0$

(C) $x - 3 = 0$ (D) $y - 3 = 0$

Q2. The equation of the latus rectum of the parabola $x^2 = -12y$ is-

(A) $y = 3$ (B) $x = 3$

(C) $y = -3$ (D) $x = -3$

Q3. The focal distance of a point(x_1, y_1) on the parabola $y^2 = 12x$ is-

(A) $x_1 + 3$ (B) $x_1 + 6$

(C) $y_1 + 6$ (D) $y_1 + 3$

Q4. The distance between the focus and the directrix of the parabola $x^2 = -8y$, is-

(A) 8 (B) 2

(C) 4 (D) 6

Q5. If (a, b) is the mid point of a chord passing through the vertex of the parabola $y^2 = 4x$, then-

(A) $a = 2b$ (B) $2a = b$

(C) $a^2 = 2b$ (D) $2a = b^2$

Q6. The equation of the parabola, whose focus is (3, –4) and directrix is the line $x + y - 2 = 0$, is -

(A) $x^2 + 2xy + y^2 - 8x + 20y + 46 = 0$

(B) $x^2 - 2xy + y^2 - 8x + 20y + 46 = 0$

(C) $x^2 - 2xy + y^2 + 8x - 20y + 46 = 0$

(D) None of these

Q7. Which of the following is not the equation of parabola -

(A) $4x^2 + 9y^2 - 12xy + x + 1 = 0$

(B) $4x^2 - 12xy + 9y^2 + 3x + 5 = 0$

(C) $2x^2 + y^2 - 4xy = 8$

(D) $4x^2 + 9y^2 - 12xy + x + 1 = 0$

Q8. The coordinates of a point on the parabola $y^2 = 8x$ whose focal distance is 4, are -

(A) $(1/2, \pm 2)$ (B) $(1, \pm 2\sqrt{2})$

(C) $(2, \pm 4)$ (D) none of these

Q9. The length of the latusrectum of the parabola whose focus is (3, 3) and directrix is $3x - 4y - 2 = 0$, is -

(A) 2 (B) 1

(C) 4 (D) None of these

Q10. If $2x + y + \lambda = 0$ is a focal chord of the parabola $y^2 = -8x$, then the value of λ is -

(A) –4 (B) 4

(C) 2 (D) –2

Q11. The equation

$$ax^2 + 4xy + y^2 + ax + 3y + 2 = 0$$

represents a parabola if a is equal to -

(a) –4 (B) 4

(C) 0 (D) 8

Shifted Parabola

Q12. Vertex, focus, latus rectum, length of the latus rectum and equation of directrix of the parabola $y^2 = 4x + 4y$ are-

(A) (1, 2), (0, 2), y = 0, 4, x = –2

(B) (–1, 2), (0, 2), x = 0, 4, x = –2

(C) (–1, 2), (1, 2), x = 0, 4, x = 2

(D) (–1, 2) (0, 2), y = 0, 2, y = –2

Q13. The length of the latus rectum of the parabola $x^2 - 4x - 8y + 12 = 0$ is-

(A) 4

(B) 6

(C) 8

(D) 10

Point and Parabola

Q14. For the parabola $y^2 = 8x$ point (2, 5) is-

(A) Inside the parabola

(B) Focus

(C) Outside the parabola

(D) On the parabola

Q15. With respect to the parabola $y^2 = 2x$, the points P (4, 2) and Q (1, 4) are such that

(A) P and Q both lie inside the parabola

(B) P lies inside whereas Q lies outside the parabola

(C) P lies outside whereas Q lies inside the parabola

(D) P and Q both lie outside the parabola

Parametric Coordinates of a Parabola

Q16. The parametric equation of the parabola $(y - 1)^2 = 12(x - 2)$ are-

(A) $6t, 3t^2$

(B) $2 + 3t, 4 + t^2$

(C) $2 + 3t^2, 1 + 6t$

(D) None of these

Q17. Parameter t of a point (9, –9) of the parabola $y^2 = 9x$ is-

(A) –1 (B) –2

(C) –2 (D) –4

Chord joining two points of a parabola

Q18. If t is the parameter for one end of a focal chord of the parabola $y^2 = 4ax$, then its length is

(A) $a\left(t + \frac{1}{t}\right)^2$ (B) $a\left(t - \frac{1}{t}\right)^2$

(C) $a\left(t + \frac{1}{t}\right)$ (D) $a\left(t - \frac{1}{t}\right)$

Q19. If QQ' is a double ordinate of a parabola $y^2 = 9x$, then the locus of its point of trisection is

(A) $y^2 = x$ (B) $y^2 = 3x$

(C) $y^2 = 6x$ (D) None of these

Q20. PSQ is a focal chord of a parabola whose focus is S and vertex A. PA and QA are produced to meet the directrix in R and T respectively. Then $\angle RST =$

(A) 90° (B) 60°

(C) 45° (D) 30°

Length of Chord of Parabola

Q21. Length of intercept by the line $4y = 3x - 48$ on the parabola $y^2 = 64x$ is -

(A) $\frac{9}{1600}$ (B) $\frac{1600}{9}$

(C) $\frac{160}{9}$ (D) None of these

Q22. If PSQ is the focal chord of the parabola $y^2 = 8x$ such that SP = 6. Then the length SQ is -

(A) 6 (B) 4

(C) 3 (D) None of these

Q23. The length of the chord of the parabola $x^2 = 4ay$ passing through the vertex and having slope $\tan\alpha$ is $(a > 0)$

(A) $|4a \operatorname{cosec}\alpha \cot\alpha|$ (B) $|4a \tan\alpha \sec\alpha|$

(C) $|4a \cos\alpha \cot\alpha|$ (D) $|4a \sin\alpha \tan\alpha|$

Tangents of Parabola

Q24. The line $y = mx + 1$ is a tangent to the parabola $y^2 = 4x$ if-

(A) $m = 1$ (B) $m = 4$

(C) $m = 2$ (D) $m = 3$

Q25. If the line $2x - 3y = k$ touches the parabola $y^2 = 6x$, then the value of k is-

(A) 27/4 (B) –81/4

(C) –7 (D) –27/4

Q26. The equation of the tangent to the parabola $y^2 = 4ax$ at the points (3, 2) is-

(A) $3y + x + 3 = 0$

(B) $3x + y + 3 = 0$

(C) $3x = y + 3$

(D) $3y = x + 3$

Q27. The equation of the tangent to the parabola $y^2 = 12x$ drawn at the upper end of its latus rectum is-

(A) $y = x + 3$ (B) $x + y = 3$

(C) $y = x - 3$ (D) None of these

Q28. If a tangent to the parabola $y^2 = ax$ makes an angle of 45° with x-axis, then its point of contact is-

(A) $\left(\frac{a}{4}, \frac{a}{2}\right)$ (B) $\left(\frac{a}{2}, \frac{a}{2}\right)$

(C) $\left(-\frac{a}{2}, \frac{a}{4}\right)$ (D) $\left(-\frac{a}{4}, \frac{a}{2}\right)$

Q29. If the line $y = mx + 1$ intersect the parabola $y^2 = 4x$ in two distinct points then $(m \neq 0)$

(A) $m < 1$ (B) $m > 1$

(C) $m = 1$ (D) $m < 2$

Q30. The straight line $y = mx + c$ touches the parabola $y^2 = 4a(x + a)$ if $(a > 0)$

(A) $c = am - \frac{a}{m}$ (B) $c = m - \frac{a}{m}$

(C) $c = am + \frac{a}{m}$ (D) None of these

Q31. The point of contact of the line $x - 2y - 1 = 0$ with the parabola $y^2 = 2(x - 3)$ is

(A) (5, 2) (B) (5, –2)

(C) (2, 5) (D) (5, 3)

Pair of tangents to parabola

Q32. The equation of pair of tangents drawn from (1, 4) to the parabola $y^2 = 12x$ is-

(A) $3x^2 + y^2 - 10x + 4y - 3 = 0$

(B) $3x^2 + y^2 - 10x + 4xy + 4y - 3 = 0$

(C) $3x^2 + y^2 + 10x - 4xy - 4y + 3 = 0$

(D) $x^2 + 3y^2 + 10x + 4xy - 4y - 3 = 0$

Q33. The chord of contact of tangents drawn from (2, 4) to the parabola $y^2 = 4x$ is-

(A) $2y = x - 2$ (B) $y = 2x + 2$

(C) $y = 2x - 1$ (D) $2y = x + 2$

Q34. The area of triangle made by the chord of contact and tangents drawn from point (4, 6) to the parabola $y^2 = 8x$ is-

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{1}{2}\sqrt{3}$ (D) None of these

Q35. The length of chord of contact of tangents drawn from point (–2, –1) to the parabola $y^2 = 4x$ is-

(A) $2\sqrt{2}$ (B) $3\sqrt{5}$

(C) 8 (D) None of these

Q36. The equation of a tangent to the parabola $y^2 = 8x$ is $y = x + 2$. The point on this line from which the other tangent to the parabola is perpendicular to the given tangents is-

(A) (–1, 1) (B) (0, 2)

(C) (2, 4) (D) (–2, 0)

Q37. The angle between tangents to the parabola $y^2 = 4ax$ at the points where it intersects with the line $x - y - a = 0$ is $(a > 0)$

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

Q38. Two tangents are drawn from the point (–2, –1) to the parabola $y^2 = 4x$. If α is the angle between them, then $\tan\alpha =$

(A) 3 (B) 1/3

(C) 2 (D) 1/2

Q39. If $y + 3 = m_1(x + 2)$ and $y + 3 = m_2(x + 2)$ are two tangents to the parabola $y^2 = 8x$, then -

(A) $m_1 + m_2 = 0$

(B) $m_1 m_2 = -1$

(C) $m_1 m_2 = 1$

(D) None of these

Normals to parabola

Q40. The equation of the normal to the parabola $y^2 = 4x$, which is parallel to the line $y - 2x + 5 = 0$ is -

(A) $2x + y - 12 = 0$

(B) $2x - y - 12 = 0$

(C) $x + 2y - 12 = 0$

(D) None of these

Q41. If $x + y = k$ is normal to $y^2 = 12x$, then k is-

(A) 3 (B) 9

(C) –9 (D) – 3

Q42. The condition that the line $\frac{x}{p} + \frac{y}{q} = 1$ be a normal to the parabola $y^2 = 4ax$ is -

(A) $p^3 = 2ap^2 + aq^2$

(B) $p^3 = 2aq^2 + ap^2$

(C) $q^3 = 2ap^2 + aq^2$

(D) None of these

Q43. The length of the normal chord to the parabola $y^2 = 4x$ which subtends a right angle at the vertex is :

(A) $6\sqrt{3}$ (B) $3\sqrt{3}$

(C) 2 (D) 1

Q44. If $x = my + c$ is a normal to the parabola $x^2 = 4ay$, then the value of c is. $(a > 0)$.

(A) $-2am - am^3$

(B) $2am + am^3$

(C) $-\frac{2a}{m} - \frac{a}{m^3}$

(D) $\frac{2a}{m} + \frac{a}{m^3}$

Q45. If $y = 2x + 3$ is a tangent to the parabola, $y^2 = 24x$, then its distrance from the parallel normal is

(A) $5\sqrt{5}$ (B) $10\sqrt{5}$

(C) $15\sqrt{5}$ (D) none of these

Q46. The numbers of distinct normals that can be drawn from $\left(\frac{11}{4}, \frac{1}{4}\right)$ to the parabola $y^2 = 4x$ is -

(A) 3 (B) 2

(C) 1 (D) 4

Common Tangent

Q47. The equation of the common tangent to the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is given by $(a > 0, b > 0)$.

(A) $xa^{1/3} + yb^{1/3} + a^{2/3}b^{2/3} = 0$

(B) $xb^{1/3} + ya^{1/3} + a^{2/3}b^{2/3} = 0$

(C) $xa^{1/3} + yb^{1/3} - a^{2/3}b^{2/3} = 0$

(D) None of these

Subtangent and Subnormal

Q48. The length of the subtangent to the parabola $y^2 = 16x$ at the point whose absicca is 4, is-

(A) 4 (B) 2

(C) 8 (D) None of these

Q49. If P is a point on the parabola $y^2 = 4ax$ such that the subtangent and subnormal at P are equal, then the coordinates of P are:

(A) (a, 2a) or (a, –2a)

(B) $(2a,\ 2\sqrt{2}a)$ or $\left(2a, -2\sqrt{2}a\right)$

(C) $\left(4a, -4a\right)$ or $\left(4a, 4a\right)$

(D) none of these

Chord with given Mid-Point

Q50. The mid-point of the chord $2x + y - 4 = 0$ of the parabola $y^2 = 4x$ is

(A) $\left(\frac{5}{2}, -1\right)$ (B) $\left(-1, \frac{5}{2}\right)$

(C) $\left(\frac{3}{2}, -1\right)$ (D) none of these

Q51. Locus of the middle points of a chords of the parabola $y^2 = 4x$ which are drawn through the vertex is-

(A) $x^2 + 4y^2 = 16$ (B) $y^2 = 2x$

(C) $y^2 = 8x$ (D) $x^2 = 2y$

Q52. The locus of the middle points of the focal chord of the parabola $y^2 = 4x$ is

(A) $y^2 = a(x - a)$ (B) $y^2 = 2a(x - a)$

(C) $y^2 = 4a(x - a)$ (D) none of these

Q53. The locus of the mid points of the focal chords of the parabola $y^2 = 4ax$ is another parabola whose vertex is given by $(a > 0)$.

(A) (a, 0) (B) (0, a)

(C) (–a, 0) (D) None of these

Highlights about Parabola

Q54. The latus rectum of a parabola whose focal chord is PSQ such that SP = 3 and SQ = 2 is given by-

(A) 24/5 (B) 6/5

(C) 12/5 (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Equation $x^2 - 2x - 2y + 5 = 0$ represents
(A) a circle with centre (1, 1)
(B) a parabola with vertex (1, 2)
(C) a parabola with directrix y = 5/2
(D) a parabola with directrix y = –1/3

Q2. A straight line touches the circle $x^2 + y^2 = 2a^2$ and the parabola $y^2 = 8ax$.
The equation of the line is (a > 0)
(A) y = x + 2a (B) y = – x – 2a
(C) y = x – 2a (D) None of these

Q3. The normals to the parabola $y^2 = 4ax$ from the point (5a, 2a) are (a > 0)
(A) y = x – 3a (B) y = – 2x + 12a
(C) y = – 3x + 33a (D) y = x + 3a

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph#1 (4 to 6)

From an external point P tangents are drawn to the parabola $y^2 = 4ax$, these tangents make angles θ_1 and θ_2 with the axis of the parabola.

Q4. If $\tan\theta_1 + \tan\theta_2 = b$, then point *P* lies on
(A) x = by (B) x + y = b
(C) y = bx (D) None of these

Q5. If $\tan\theta_1 \cdot \tan\theta_2 = c$, then point *P* lies on
(A) x = cy (B) cx = a
(C) cy = a (D) None of these

Q6. If $\cot\theta_1 + \cot\theta_2 = d$, then point *P* lies on
(A) x = dy (B) y = dx
(C) y = ad (D) None of these

Paragraph#2 (7 to 9)

P is intersecting point of tangents at two points A and B on the parabola $y^2 = 4ax$. If angle between these tangents is θ, then locus of *P*, if

Q7. $\theta = 45^\circ$
(A) x = –a
(B) $y^2 - 4ax = (x + a)^2$
(C) $y^2 - 3x^2 - 10ax - 3a^2 = 0$
(D) None of these

Q8. $\theta = 60^\circ$
(A) $y^2 - 3x^2 - 10ax - 3a^2 = 0$
(B) $y^2 + 3x^2 - 10ax - 3a^2 = 0$
(C) $x^2 - 3y^2 - 10ay - 3a^2 = 0$
(D) $x^2 + 3y^2 - 10ay - 3a^2 = 0$

Q9. $\theta = 90^\circ$
(A) x = a (B) x = –a
(C) y = a (D) y = –a

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q10. The parabola $y^2 = 4ax$ has a chord AB joining points $A(at_1^2, 2at_1), B(at_2^2, 2at_2)$.

Match List - I with List - II.

List - I	**List - II**
(P) If normal chord AB subtend 90° at origin, then	(1) $t_2 = -t_1 + 2$
(Q) If AB is a focal chord then	(2) $t_2 = \frac{-4}{t_1}$
(R) If AB subtend 90° at point (0, 0) then	(3) $t_2 = \frac{-1}{t_1}$
(S) If AB if inclined at 45° to axis of parabola then	(4) $t_2 = -t_1 - \frac{2}{t_1}$

Code :

(A) P-4, Q-3, R-1, S-2

(B) P-4, Q-3, R-2, S-1

(C) P-1, Q-3, R-4, S-2

(D) P-3, Q-4, R-2, S-4

SUBJECTIVE TYPE

Q11. If one extremity of the focal chord of the parabola $y^2 = 16x$ is (1, –4) then find other extremity.

Q12. Find the length of a chord of a parabola $y^2 = 4x$ which passes through vertex and makes an angle of 45° with x-axis.

Q13. Find the locus of perpendicular tangents of the parabola $x^2 - 4x + 2y + 3 = 0$.

Q14. Find the common tangent of the parabola $x^2 = 4ay$ and $y^2 = 4ax (m > 0)$.

Q15. Find the equation of a chord of the parabola $y^2 = 6x$ if mid point of the chord is (–1, 1).

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The axis of a parabola is along the line $y = x$ and the distance of its vertex from origin is $\sqrt{2}$ and that of origin from its focus is $2\sqrt{2}$. If vertex and focus both lie in the first quadrant, then the equation of the parabola is -

(A) $(x + y)^2 = (x - y - 2)$

(B) $(x - y)^2 = (x + y - 2)$

(C) $(x - y)^2 = 4(x + y - 2)$

(D) $(x - y)^2 = 8(x + y - 2)$

Q2. A parabola is drawn with focus at (3, 4) and vertex at the focus of the parabola $y^2 - 12x - 4y + 4 = 0$. The equation of the parabola is

(A) $x^2 - 6x - 8y + 25 = 0$

(B) $y^2 - 8x - 6y + 25 = 0$

(C) $x^2 - 6x + 8y - 25 = 0$

(D) $x^2 + 6x - 8y - 25 = 0$

Q3. The mirror image of the directrix of the parabola $y^2 = 4(x + 1)$ in the line mirror $x + 2y = 3$ is -

(A) $x = -2$

(B) $4y - 3x = 16$

(C) $x - 3y = 0$

(D) $x + y = 0$

Q4. The locus of the vertices of the family of parabolas $y = \frac{a^3x^2}{3} + \frac{a^2x}{2} - 2a$ is-

(A) $xy = \frac{3}{4}$ (B) $xy = \frac{35}{16}$

(C) $xy = \frac{64}{105}$ (D) $xy = \frac{105}{64}$

Q5. The point $(-2m, m + 1)$ is an interior point of the smaller region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 4x$. Then m belongs to the interval :

(A) $-5 - 2\sqrt{6} < m < 1$

(B) $0 < m < 4$

(C) $-1 < m < \frac{3}{5}$

(D) $-1 < m < -5 + 2\sqrt{6}$

Q6. If the focus of parabola $(y - k)^2 = 4(x - h)$ always lies between the line $x + y = 1$ and $x + y = 3$ then

(A) $0 < h + k < 2$

(B) $0 < h + k < 1$

(C) $1 < h + k < 2$

(D) $1 < h + k < 3$

Q7. The number of points with integral coordinates that lie in the interior of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 4x$ is -

(A) 8 (B) 10

(C) 16 (D) None of these

Q8. The curve described parametrically by $x = t^2 + t + 1$, $y = t^2 - t + 1$ represents-

(A) a pair of straight line

(B) an ellipse

(C) a parabola

(D) a hyperbola

Q9. AB is a chord of the parabola $y^2 = 4ax$ with vertex at A, BC is drawn perpendicular to AB meeting the axis at C. The projection of BC on the x-axis is (a > 0)

(A) a (B) 2a

(C) 4a (D) 8a

Q10. Coordinates of any point on the parabola, whose focus is $\left(\frac{-3}{2}, -3\right)$ and the directrix is $2x + 5 = 0$ is given by

(A) $(2t^2 + 2, 2t - 3)$ (B) $(2t^2 - 2, 2t - 3)$

(C) $(2t^2 - 2, 2t + 3)$ (D) None of these

Q11. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabolas $y^2 = 4ax$ and $x^2 = 4ay$, then-

(A) $d^2 + (2b + 3c)^2 = 0$

(B) $d^2 + (3b + 2c)^2 = 0$

(C) $d^2 + (2b - 3c)^2 = 0$

(D) $d^2 + (3b - 2c)^2 = 0$

Q12. A line L passing through the focus of the parabola $y^2 = 4(x-1)$ intersects the parabola in two distinct points. If 'm' be the slope of the line L, then

(A) $m \in (-1, 1)$

(B) $m \in (-\infty, -1) \cup (1, \infty)$

(C) $m \in R$

(D) $m \in R - \{0\}$

Q13. The straight line $y = mx + c$ $(m > 0)$ touches the parabola $y^2 = 8(x + 2)$ then the minimum value taken by c is

(A) 4 (B) 8

(C) 12 (D) 6

Q14. If the tangent at P on $y^2 = 4ax$ meets the tangent at the vertex in Q and S in the focus of the parabola, then $\angle SQP$ is equal to :

(A) $\pi/3$ (B) $\pi/4$

(C) $\pi/2$ (D) $2\pi/3$

Q15. From an external point P pair of tangents are drawn to the parabola $y^2 = 4x$. If θ_1 and θ_2 are inclination of their tangents with the axis of x such that $\theta_1 + \theta_2 = \frac{\pi}{4}$ then the locus of P is-

(A) $x - y + 1 = 0$

(B) $x + y - 1 = 0$

(C) $x - y - 1 = 0$

(D) $x + y + 1 = 0$

Q16. Angle between the tangents drawn from (1, 4) to the parabola $y^2 = 4x$ is-

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Q17. The tangents at three points A, B, C on the parabola $y^2 = 4x$, taken in pairs intersect at the points P, Q and R. If Δ, Δ' be the areas of the triangles ABC and PQR respectively, then -

(A) $\Delta = 2\Delta'$ (B) $\Delta' = 2\Delta$

(C) $\Delta = \Delta'$ (D) None of these

Q18. The condition that the straight line $\ell x + my + n = 0$ touches the parabola $x^2 = 4ay$ is -

(A) $bn = am^2$ (B) $a\ell^2 - mn = 0$

(C) $\ell n = am^2$ (D) $am = \ell n^2$

Q19. The set of points on the axis of the parabola $y^2 - 4x - 2y + 5 = 0$ from which all the three normals to the parabola are real is

(A) (k, 0) ; k > 1

(B) (k, 1) ; k > 3

(C) (k, 2) ; k > 6

(D) (k, 3) ; k > 8

Q20. The condition that the parabolas $y^2 = 4c(x-d)$ and $y^2 = 4ax$ have a common normal other than x-axis $(a > c > 0)$ is-

(A) $2a < 2c + d$ (B) $2c < 2a + d$

(C) $2d < 2a + c$ (D) $2d < 2c + a$

Q21. If P(2, –4) and Q are points on the parabola $y^2 = 8x$ and the chord PQ subtends a right angle at the vertex of the parabola, then the coordinates of the point of intersection of normal at P and Q is

(A) (30, 24) (B) (30, –24)

(C) (24, 30) (D) (24, –30)

Q22. The shortest distance between the parabolas $y^2 = 4x$ and $y^2 = 2x - 6$ is

(A) 2 (B) $\sqrt{5}$

(C) 3 (D) None of these

Q23. Tangent to the curve $y = x^2 + 6$ at point P (1, 7) touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q. Then coordinate of Q is -

(A) (–6, 11) (B) (6, –11)

(C) (–6, –7) (D) (–6, –11)

Q24. Consider a circle with centre lying on the focus of the parabola $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle and the parabola is-

(A) (p/2, p) (B) (– p/2, p)

(C) (– p/2, – p) (D) None of these

Q25. Above x-axis, the equation of the common tangents to the circle $(x - 3)^2 + y^2 = 9$ and parabola $y^2 = 4x$ is-

(A) $\sqrt{3}\,y = 3x + 1$

(B) $\sqrt{3}\,y = -(x + 3)$

(C) $\sqrt{3}\,y = x + 3$

(D) $\sqrt{3}\,y = -(3x + 1)$

Q26. The locus of point of intersection of tangents to the parabolas $y^2 = 4(x+1)$ and $y^2 = 8(x+2)$ which are perpenidcular to each other is -

(A) $x + 7 = 0$ (B) $x - y = 4$

(C) $x + 3 = 0$ (D) $y - x = 12$

Q27. The coordinates of the point on the parabola $y^2 = 8x$, which is at minimum distance from the circle $x^2 + (y+6)^2 = 1$, are -

(A) (2, –4) (B) (18, –12)

(C) (2, 4) (D) none of these

Q28. If the tangent at the point P(2, 4) to the parabola $y^2 = 8x$ meets the parabola $y^2 = 8x + 5$ at Q and R, then the mid-point of QR is -

(A) (2, 4) (B) (4, 2)

(C) (7, 9) (D) None of these

Q29. The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix

(A) $x = -a$ (B) $x = -a/2$

(C) $x = 0$ (D) $x = a/2$

Q30. A ray of light moving parallel to the x-axis gets reflected from a parabolic mirror whose equation is $(y-2)^2 = 4(x+1)$. After reflection, the ray must pass through the point

(A) (–2, 0) (B) (–1, 2)

(C) (0, 2) (D) (2, 0)

Q31. If the tangent to the parabola $y^2 = 4ax$ meets the axis in T and tangent at the vertex A in Y and the rectangle TAYG is completed, then the locus of G is $(a > 0)$

(A) $y^2 + 2ax = 0$

(B) $y^2 + ax = 0$

(C) $x^2 + ay = 0$

(D) None of these

Q32. If the tangents and normals at the extremities of a focal chord of a parabola intersect at (x_1, y_1) and (x_2, y_2) respectively, then -

(A) $x_1 = x_2$

(B) $x_1 = y_2$

(C) $y_1 = y_2$

(D) $x_2 = y_1$

Q33. If M is the foot of the perpendicular from a point P on a parabola $y^2 = 4ax$ to its directrix and SPM is an equilateral triangle, where S is the focus, then SP is equal to $(a > 0)$

(A) a (B) 2a (C) 3a (D) 4a

Q34. A line bisecting the ordinate PN of a point $P\left(at^2, 2at\right), t > 0,$ on the parabola $y^2 = 4ax$ is drawn parallel to the axis to meet the curve at Q. If NQ meets the tangent at the vertex at the point T, then the coordinates of T are

(A) $\left(0, \frac{4}{3}at\right)$ (B) $(0, 2at)$

(C) $\left(\frac{1}{4}at^2, at\right)$ (D) $(0, at)$

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. P is a point which moves in the x-y plane such that the point P is nearer to the centre of a square than any of the sides. The four vertices of the square are $(\pm a, \pm a)$. The region in which P will move is bounded by pairs of parabolas of which one has the equation.

(A) $y^2 = a^2 + 2ax$

(B) $x^2 = a^2 + 2ay$

(C) $y^2 + 2ax = a^2$

(D) $x^2 = a^2 - ay$

Q2. The ends of a line segment are P (1, 3) and Q (1, 1) . R is a point on the line segment PQ such that PR : QR = $1:\lambda$. If R is an interior point of the parabola $y^2 = 4x$, then

(A) $\lambda \in (0,1)$ (B) $\lambda \in \left(-\frac{3}{5}, 1\right)$

(C) $\lambda \in \left(\frac{1}{2}, \frac{3}{5}\right)$ (D) none of these

Q3. Let PQ be a chord of the parabola $y^2 = 4x$. A circle drawn with PQ as a diameter passes through the vertex V of the parabola. If area of $\Delta PVQ = 20$ sq. unit, then coordinates of P are

(A) (–16, –8) (B) (–16, 8)

(C) (16, –8) (D) (16, 8)

Q4. The equation of a tangent to the parabola $y^2 = 8x$ which makes an angle 45° with the line $y = 3x + 5$ is -

(A) $2x + y + 1 = 0$ (B) $y = 2x + 1$

(C) $x - 2y + 8 = 0$ (D) $x + 2y - 8 = 0$

Q5. A tangent to a parabola $y^2 = 4ax$ is inclined at $\frac{\pi}{3}$ with the axis of the parabola. The point of contact is -

(A) $(a/3, 2a/\sqrt{3})$

(B) $(3a, -2\sqrt{3}a)$

(C) $(3a, 2\sqrt{3}a)$

(D) $(a/3, -2a/\sqrt{3})$

Q6. The equation of the tangent to the parabola $y^2 = 9x$ which goes through the point (4, 10) is -

(A) $x + 4y + 1 = 0$

(B) $9x + 4y + 4 = 0$

(C) $x - 4y + 36 = 0$

(D) $9x - 4y + 4 = 0$

Q7. Which of the following is (are) true about the parabola $y^2 = 4ax$ $(a > 0)$?

(A) If t_1, t_2 are end points of focal chord, then $t_1t_2 = -1$.

(B) Tangent at the end of a focal chord cuts at right angle at directrix.

(C) Distance of any point on the parabola from directrix is equal to the sum of 'a' and abscissa of the point.

(D) End points of latusrectum are (a, 2a), (–a, 2a).

Q8. The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex if

(A) $m = 1$

(B) $m = \sqrt{2}$

(C) $m = -\sqrt{2}$

(D) $m = 1/\sqrt{2}$

Q9. The equations of the common tangents to the parabola $y = x^2$ and $y = -x^2 + 4x - 4$ is/ are-

(A) $y = 4(x - 1)$

(B) $y = 0$

(C) $y = -4(x - 1)$

(D) $y = -30x - 50$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 10 to 12)

The straight line $y + tx = 2at + at^3$ is a normal to the parabola $y^2 = 4ax$ for all real value of t and foot of the normal is given by $(at^2, 2at)$.

Q10. Number of real normals that can be drawn from the (4, 0) to the parabola $y^2 = 16x$ is

(A) 1 (B) 2

(C) 3 (D) None of these

Q11. If the ordinates of the points P and Q on the parabola $y^2 = 12x$ are in the ratio 1 : 2 then the locus of the point of intersection of normals to the parabola at P and Q is

(A) $343y^2 = 12(x - 6)^3$

(B) $343y^2 = 12(x + 6)^3$

(C) $343y^2 = -12(x - 6)^3$

(D) None of these

Q12. A point through which three normals of parabola $y^2 = 4ax$ are passing, two of which are making angles α and β with x-axis, where $\tan\alpha \tan\beta = 2$, lies on the curve

(A) $y(y^2 - 2ax) = 0$

(B) $y(y^2 + 2ax) = 0$

(C) $y(y^2 - 4ax) = 0$

(D) $y(y^2 - ax) = 0$

Paragraph # 2 (Ques. 13 to 15)

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

Q13. The ratio of the areas of the triangle PQS and PQR is :-

(A) $1:\sqrt{2}$ (B) 1 : 2 (C) 1 : 4 (D) 1 : 8

Q14. The radius of the circumcircle of the triangle PRS is :-

(A) 2 (B) $3\sqrt{3}$

(C) $3\sqrt{2}$ (D) $2\sqrt{3}$

Q15. The radius of the incircle of the triangle PQR is :-

(A) 4 (B) 3

(C) $\frac{8}{3}$ (D) 2

MATCH THE COLUMN TYPE

Following questions contain statements given in two columns, which have to be matched. The statements in ***Column-I*** *are labelled as A, B, C and D while the statements in* ***Column-II*** *are labelled as (P), (Q), (R), (S) and (T). Any given statement in* ***Column-I*** *can have correct matching with* ***ONE OR MORE*** *statement(s) in* ***Column-II****.*

Q16.

Column-I	**Column-II**
(A) The equation of the common normals to the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is/are	(P) $x + a = 0$
(B) A pair of tangents drawn from a point P to the parabola $y^2 = 4ax$ intersects the coordinate axes in concylic points. The locus of P is	(Q) $x = a$
(C) The locus of point from which tangents drawn to parabolas $y^2 = 4a(x + a)$ and $y^2 = 8a(x + 2a)$ are mutually perpendicular is	(R) $x + y = 3a$
(D) The chord of contact of a point P with respect to the parabola $y^2 - 4ax = 0$ subtends right angle at the vertex. The locus of P is	(S) $x + 3a = 0$
	(T) $x = 4a$

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	A	C	D	B	C	C	A	B
11	12	13	14	15	16	17	18	19	20
B	B	C	C	B	C	B	A	A	A
21	22	23	24	25	26	27	28	29	30
B	C	B	A	D	D	A	A	A	C
31	32	33	34	35	36	37	38	39	40
A	C	D	B	B	D	D	A	B	B
41	42	43	44	45	46	47	48	49	50
B	A	A	A	C	B	A	C	A	A
51	52	53	54						
B	B	A	A						

Ex - 1A

1. (B, C) **2.** (A, B) **3.** (A, B) **4.** (C) **5.** (B)

6. (C) **7.** (B) **8.** (A) **9.** (B) **10.** (B)

11. (16, 16) **12.** $4\sqrt{2}$ **13.** $y = 1$ **14.** $x + y + a = 0$ **15.** $y - 3x = 4$

Ex-2									
1	2	3	4	5	6	7	8	9	10
D	A	B	D	D	A	D	C	C	B
11	12	13	14	15	16	17	18	19	20
A	D	A	C	C	B	A	B	B	A
21	22	23	24	25	26	27	28	29	30
A	B	C	A	C	C	A	A	C	C
31	32	33	34						
B	C	D	A						

Ex - 2A

1. (A, B, C) **2.** (A, C) **3.** (C, D) **4.** (A, C) **5.** (A, D)

6. (C, D) **7.** (A, B, C) **8.** (B, C) **9.** (A,B) **10.** (A)

11. (A) **12.** (C) **13.** (C) **14.** (B) **15.** (D)

16. [A-R; B-Q; C-S; D-T]